

23 A uniform metal wire of length L and diameter d has spring constant k .

What is the Young modulus of the metal?

A $\frac{\pi d^2}{4kL}$

B $\frac{\pi d^2}{kL}$

C $\frac{kL}{\pi d^2}$

D $\frac{4kL}{\pi d^2}$

24 An aircraft produces a sound at a frequency of 30.0 Hz